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# Get Started with Kubernetes Ultimate Hands-on Labs and Tutorials

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## Get Started with Kubernetes | Ultimate Hands-on Labs and Tutorials

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A Curated List of Kubernetes Labs and Tutorials

- A \$0 Learning Platform for All Levels - from the ground Up
- Over 500+ Highly Interactive Docker Tutorials and Guides
- Well tested on Kubernetes Cluster and can be run on Browser (no Infrastructure required)

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## Featured Articles

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- [Top 5 Alert and Monitoring Tools for Kubernetes](#)
- [Top 5 Machine Learning Tools For Kubernetes](#)
- [Top 5 Cluster Management Tools for Kubernetes in 2023](#)
- [10 Tips for Right Sizing Your Kubernetes Cluster](#)
- [Step-by-Step Guide to Deploying and Managing Redis on Kubernetes](#)
- [Update Your Kubernetes App Configuration Dynamically using ConfigMap](#)
- [Streamline Your Deployment Workflow: Utilizing Docker Desktop for Local Development and OpenShift for Production Deployment](#)
- [The Impact of Kube-proxy Downtime on Kubernetes Clusters](#)
- [How to add a Secret to a Deployment in Kubernetes using Kubectl patch](#)

## Pre-requisite:

- [Introductory Slides](#)
- [Deep Dive into Kubernetes Architecture](#)

## Preparing 5-Node Kubernetes Cluster

### PWK:

- [Preparing 5-Node Kubernetes Cluster](#)
- [Setting up WeaveScope For Visualization on Kubernetes](#)
- [Running Portainer on 5 Node Kubernetes Cluster](#)

### GKE

- [Setting up GKE Cluster](#)
- [Setting up Weavescope for Visualization on GKE](#)

## Docker Desktop for Mac

- [Setting up Kubernetes Cluster on AWS using Kops running on Docker Desktop for Mac](#)

## Ubuntu

- [Setting up Kubernetes on Ubuntu](#)

## Using Kubectl

- [Kubectl for Docker Beginners](#)
- [Accessing Kubernetes API](#)
- [How to know if Kubernetes is using Docker or Containerd as a runtime](#)

## Kubernetes CRUD

- [Using Python](#)
- [Using Go](#)

## Using AI

- [Deploying Kubernetes via AI](#)

## Pods101

- [Introductory Slides](#)
- [Deploying Your First Nginx Pod](#)
- [Viewing Your Pod](#)
- [Where is your Pod running on?](#)
- [Pod Output in JSON](#)
- [Executing Commands against Pod](#)
- [Terminating a Pod](#)
- [Adding a 2nd container to a Pod](#)
- [Labels and Selectors in a Pod](#)

## Kubernetes Tools for Pods

- [Kubetail](#)

## ReplicaSet101

- [Introductory Slides](#)
- [Creating Your First ReplicaSet - 4 Pods serving Nginx](#)
- [Removing a Pod from ReplicaSet](#)
- [Scaling & Autoscaling a ReplicaSet](#)
- [Best Practices](#)
- [Deleting ReplicaSets](#)

## Deployment101

- [Introductory Slides](#)
- [Creating Your First Deployment](#)
- [Checking the list of application deployment](#)
- [Scale up/down application deployment](#)
- [Scaling the service to 2 Replicas](#)

- [Perform rolling updates to application deployment](#)
- [Rollback updates to application deployment](#)
- [Cleaning Up](#)

## ConfigMaps101

- [What are ConfigMaps?](#)

## Scheduler101

- [How Kubernetes Selects the Right node?](#)
- [Node Affinity](#)
- [Anti-Node Affinity](#)
- [Nodes taints and tolerations](#)

## Services101

- [Introductory Slides](#)
- [Deploy a Kubernetes Service?](#)
- [Labels and Selectors](#)
- [Service Exposing More Than One Port](#)
- [Kubernetes Service Without Pods?](#)
- [Service Discovery](#)
- [Connectivity Methods](#)
- [Headless Service In Kubernetes?](#)

## StatefulSets101

- [The difference between a Statefulset and a Deployment](#)
- [Deploying a Stateful Application Using Kubernetes Statefulset?](#)
- [Deploying NFS Server](#)
- [Deploying PV](#)
- [Deploying PVC](#)
- [Using Volume](#)
- [Recreate Pod](#)

## DaemonSet101

- [Why DaemonSets in Kubernetes?](#)
- [Creating your first DaemonSet Deployment](#)
- [Restrict DaemonSets To Run On Specific Nodes](#)
- [How To Reach a DaemonSet Pod](#)

## Jobs101

- [Creating Your First Kubernetes Job](#)
- [Multiple Parallel Jobs \(Work Queue\)](#)

## Ingress101

- [What is Kubernetes ingress?](#)
  - [NodePort](#)
  - [Load Balancer](#)
  - [Ingress](#)
  - [How to Use Nginx Ingress Controller](#)
  - [Ingress Controllers and Ingress Resources](#)
- [Ingress with EKS](#)

## RBAC101

- [Role-Based Access Control \(RBAC\) Overview](#)
- [Creating a Kubernetes User Account Using X509 Client Certificate](#)

## Service Catalog101

- [What is Kubernetes Service Catalog?](#)
- [Creating a sample Service Catalog](#)
- [Installing Service Catalog Helm Chart](#)

## Cluster Networking101

- [What Is Cluster Networking In Kubernetes Sense?](#)
- [Kubernetes Networking Rules](#)
- [Types of Networks](#)
  - [Underlay Network](#)
  - [Overlay Network](#)
- [What is a Container Network Interface \(CNI\)?](#)
  - [AWS VPC CNI for Kubernetes](#)
  - [AZURE CNI for Kubernetes](#)
  - [Calico](#)
  - [Cilium](#)
  - [Weave Net from WeaveWorks](#)
  - [Flannel](#)
- [LAB- Weave Net Implementation](#)

## Network Policies101

- [What is a Kubernetes Network Policy?](#)
- [Creating Your First NetworkPolicy Definition](#)
- [How can we fine-tune Network Policy using selectors?](#)
- [Deny Ingress Traffic That Has No Rules](#)
- [Deny Egress Traffic That Has No Rules](#)
- [Allow All Ingress Traffic Exclusively](#)
- [Allow All Egress Traffic Exclusively](#)

## Monitoring101

- [Monitoring in Kubernetes](#)
- [Core Monitoring Pipeline](#)
- [Services Monitoring Pipeline](#)
- [What should you consider in Kubernetes Services Pipeline?](#)
- [What about Metrics Visualization?](#)
- [Changes To Watch For](#)
  - [Heapster is Going Away](#)
  - [Metrics Server Will Get More Cool Features](#)

## Logging101

- [Logging introduction](#)
- [Elasticsearch](#)
- [Fluentd](#)
- [Fluentd on Kubernetes](#)
- [Fluent Bit](#)
- [ELK on Kubernetes](#)

## Autoscalers101

- [What are autoscalers](#)
- [Autoscaler lab](#)

## Helm101

- [What is Helm?](#)
- [Installing a Helm Chart](#)
- [Helm Charts](#)
- [Helm Chart Hooks](#)
- [Helm Chart Testing](#)
- [Helm Chart Repository](#)

## AKS101

- [What is AKS?](#)
- [AKS Networking](#)
- [AKS IAM](#)
- [AKS Storage](#)
- [AKS Service Mesh](#)
- [AKS KEDA](#)

## Security101

- [What is DevSecOps?](#)
- [Securing your cluster](#)

## GitOps101

- [What is GitOps](#)
- [ArgoCD](#)
- [ArgoCD with EKS](#)

## Managed Kubernetes Service

- [Managed Kubernetes Service Intro](#)

## EKS101

- [What is EKS](#)

## LKE101

- [What is LKE](#)

## GKE101

- [What is GKE](#)
- [Google Cloud Run](#)
- [GKE Service Mesh](#)

## Loft101

- [What is Loft](#)
- [Loft Lab](#)

## Shipa101

- [What is Shipa](#)
- [Shipa Lab](#)

## DevSpace101

- [What is DevSpace](#)
- [DevSpace Lab](#)

## KubeSphere101

- [What is KubeSphere](#)
- [KubeSphere lab](#)

## Kubernetes with GitLab 101

- [What is GitLab](#)
- [Kubernetes with GitLab](#)
- [GitLab runner on Kubernetes](#)

## Kubernetes with Jenkins

- [Jenkins on Kubernetes](#)
- [Using Jenkins on Kubernetes](#)

## Strimzi (Kafka on Kubernetes)

- [What is Kafka](#)
- [Running Kafka on Kubernetes](#)

## Java client for Kubernetes

- [Introduction](#)

## KEDA

- [What is KEDA](#)
- [KEDA lab](#)

## Terraform EKS

- [What is Terraform](#)
- [Terraform EKS Lab](#)

## Disaster Recover

- [What is Disaster Recovery](#)
- [DR Lab](#)

## For Node Developers

- [Kubernetes for Node Developers](#)

## Cheat Sheets

- [Kubernetes Cheat Sheet](#)
- [Helm Cheat Sheet](#)

## Contributors

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- [Saiyam Pathak](#)
- [Divyajeet Singh](#)
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## Workshop Video



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## Contribution Guidelines

### Step 1. Clone the repository

```
git clone https://github.com/collabnix/kubelabs
```

### Step 2. Add \_config\_dev.yml

Add the following entry for local access

```
url: http://127.0.0.1:4000
```

### Step 3. Run the container

```
docker run --rm \
-v "$PWD:/srv/jekyll" \
-e BUNDLE_PATH="/srv/jekyll/.bundles_cache" \
-p 4000:4000 \
jekyll/builder:3.8 \
bash -c "gem install bundler && bundle install && bundle exec jekyll serve --host 0.0.0.0 --verbose -
```

## Further References:

- [Kubetools](#)

[Next: Kubernetes201](#)

Launch Terminal

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